Algebraic Geometry II

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Contents

In Algebraic Geometry I, we studied the classical approach of varieties in affine or projective space. While this is very geometric, there remain issues: Firstly, one needs the base field K to be algebraically closed: otherwise, Hilbert's Nullstellensatz fails, and the correspondence between geometric and algebraic objects breaks down (e.g. $X^2 + 1 = 0$ has no solutions in $\mathbb{R}$). In fact, we would like to not only look at non-algebraically closed fields, but more generally at algebraic sets over arbitrary rings, to study, for instance, algebraic equations over the integers using geometric methods. Lecture 1 Apr 13, 2026

Another issue of classical varieties comes from intersections: for example, take $K = \mathbb{C}$ and look at $V(y - x^2) \cap V(y) \subseteq \mathbb{A}^2$. One can compute this intersection to be $V(x^2, y) = V(x, y)$, which corresponds to the single point $(0, 0)$. However, we would like to distinguish the algebraic sets $V(x^2, y)$ and $V(x, y)$, since the former carries the extra information of "multiplicity". Algebraically, this corresponds to non-reduced K-algebras. If we had such a notion of multiplicity, we could prove more powerful theorems. For instance, Bézout's theorem simply reads $|X \cap Y| = \deg X \deg Y$, where the points in $X \cap Y$ are counted with multiplicity.

To resolve these problems, and at the same time introduce a much more general and powerful framework, Grothendieck introduced the language of schemes. In the classical setting, note that the points of an affine variety $X \subseteq \mathbb{A}^n$ correspond to the maximal ideals of $A(X)$. In the new setting, we will look at all prime ideals of $A(X)$.

For an arbitrary ring A, we can give $\operatorname{Spec} A$ a Zariski topology, and define regular functions on open subsets of $\operatorname{Spec} A$. This is all similar to the classical ideas, but involves more technical objects. This will defie the notion of an *affine scheme*.

1 Sheaves

Definition 1.1. Let X be a topological space. A *presheaf* $\mathcal{F}$ of abelian groups on X consists of an abelian group $\mathcal{F}(U)$ for each open subset $U \subseteq X$ and a group homomorphism $\rho_{U,V} : \mathcal{F}(U) \to \mathcal{F}(V)$ for any $V \subseteq U$ open, such that $\rho_{U,U} = \mathrm{id}_{\mathcal{F}(U)}$ and $\rho_{V,W} \circ \rho_{U,V} = \rho_{U,W}$ for any $W \subseteq V \subseteq U$ open.

In other words, a presheaf is a contravariant functor from the category of open subsets of X to the category of abelian groups. The homomorphisms $\rho_{U,V}$ are called *restriction maps*. An element $s \in \mathcal{F}(U)$ is called a section of $\mathcal{F}$ on U. If $V \subseteq U$ is open, we will usually write $s|_V := \rho_{UV}(s)$.

Remark 1.2. Analogously one defines presheaves of sets, groups, rings, R-modules etc.

Example 1.3. Let X be a topological space, and let $\mathcal{F}(U) := \{f : U \to \mathbb{R} \mid f \text{ continuous}\}$, with ρ_{UV} the usual restriction of functions. Similarly, if X carries additional structure, one can look at conformal, differentiable or holomorphic functions on X.

Let X be a quasi-projective variety over a field K. Then the regular functions on X form a presheaf $\mathcal{O}_X(U) = \{f : U \to X \mid f \text{ regular}\}$.

In all examples, the presheaf $\mathcal{F}$ satisfies the following two properties:

Definition 1.4. A presheaf $\mathcal{F}$ on X is a *sheaf* if the following two conditions hold for any open $U \subseteq X$ and every open cover $U = \bigcup_{i \in I} U_i$:

(i) If $s, t \in \mathcal{F}(U)$ with $s|_{U_i} = t|_{U_i}$ for all $i \in I$, then $s = t$.

(ii) If $s_i \in \mathcal{F}(U_i)$ for all $i \in I$ with $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$ for all $i, j \in I$, then there exists $s \in \mathcal{F}(U)$ with $s|_{U_i} = s_i$ for all $i \in I$.

In fact, the section s in (ii) is unique by (i).

Remark 1.5. The examples in 1.3 are all sheaves. For an example of a presheaf that is not necessarily a sheaf, let $1 \neq A$ be any abelian group, X a topological space, and $\mathcal{F}(U) := A$ for all opens $U \subseteq X$. If X is not connected, this is not a sheaf. To rectify this, give A the discrete topology, and define the *constant sheaf* $\underline{A}$ by

$$\underline{A}(U) := \operatorname{Hom}(U, A) = \{f : U \to A \mid f \text{ locally constant}\}$$

Definition 1.6. Let $\mathcal{F}$ be a presheaf on X, and let $x \in X$. Then the *stalk* of $\mathcal{F}$ at x is defined as the direct limit

$$\mathcal{F}_x := \varinjlim_{x \in U} \mathcal{F}(U) = \{(U, s) \mid x \in U, s \in \mathcal{F}(U)\} / \sim,$$

where $(U, s) \sim (V, t)$ if there exists an open $x \in W \subseteq U \cap V$ with $s|_W = t|_W$.

Remark 1.7. $\mathcal{F}_x$ is an abelian group with operation $[(U, s)] + [(V, t)] = [(U \cap V, s|_{U\cap V} + t|_{U\cap V})]$.

Definition 1.8. Let X be a topological space. A morphism $\varphi : \mathcal{F} \to \mathcal{G}$ of (pre)sheaves on X is a family of morphisms $\varphi_U : \mathcal{F}(U) \to \mathcal{G}(U)$ for all opens $U \subseteq X$, such that for all opens $V \subseteq U$, the following diagram commutes:

$$\begin{tikzcd} \mathcal{F}(U) \arrow[r, "\varphi_U"] \arrow[d, "\rho_{UV}"] & \mathcal{G}(U) \arrow[d, "\rho_{UV}"] \\ \mathcal{F}(V) \arrow[r, "\varphi_V"] & \mathcal{G}(V) \end{tikzcd}$$

φ is an isomorphism if there exists an inverse morphism $\psi : \mathcal{G} \to \mathcal{F}$.

Remark 1.9. In other words, a morphism of (pre)sheaves is a natural transformation of the two functors.

Lecture 2
Apr 16, 2026

Proposition 1.10. *Let $x \in X$. A morphism $\varphi : \mathcal{F} \to \mathcal{G}$ of presheves on X induces a morphism $\varphi_x : \mathcal{F}_x \to \mathcal{G}_x$ of stalks, given explicitly by $[(U, f)] \mapsto [(U, \varphi(U)(f))]$.*

Proof. This follows directly from the universal property of direct limits. $\square$

Proposition 1.11. *Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of schemes. Then φ is an isomorphism if and only if $\varphi_x : \mathcal{F}_x \to \mathcal{G}_x$ is an isomorphism for every $x \in X$.*

Note that this fails for presheaves, and hence gives some motivation for the conditions in the definition of a sheaf.

Proof. If φ is an isomorphism, then so is φ_x (more precisely $(\varphi_x)^{-1} = (\varphi_x)^{-1}$) by functoriality of direct limits.

Conversely, assume that φ_x is an isomorphism for every $x \in X$, and let $U \subseteq X$ be open. Assume $s \in \ker \varphi_U$. For any $x \in U$, we have $\varphi_x([(U, s)]) = [U, \varphi(U)(s)] = 0$, so by injectivity of φ_x we get $[(U, s)] = 0$ in $\mathcal{F}_x$. That is, $s = 0$ on some neighbourhood $x \in U_x \subseteq U$. Then $U = \bigcup_{x\in U} U_x$, so by 1.4(1) we get $s = 0 \in \mathcal{F}(U)$. Hence φ_U is injective.

For surjectivity, let $t \in \mathcal{G}(U)$. For any $x \in U$, φ_x is surjective, so let $[(U_x, s_x)] \in \mathcal{F}_x$ be a preimage of $[(U, t)] \in \mathcal{G}_x$, wlog $U_x \subseteq U$. Then $U = \bigcup_{x\in X} U_x$ and $s_x \in \mathcal{F}(U_x)$. To check the gluing condition, let $x, y \in U$ and compute

$$\varphi_{U_x\cap U_y}(s_x|_{U_x\cap U_y}) \overset{1.8}{=} (\varphi_{U_x}(s_x))|_{U_x\cap U_y} = t|_{U_x}|_{U_x\cap U_y} \overset{1.1}{=} t|_{U_x\cap U_y} = \varphi_{U_x\cap U_y}(s_y|_{U_x\cap U_y}),$$

thus by injectivity $s_x|_{U_x \cap U_y} = s_y|_{U_x \cap U_y}$. Hence by 1.4(2) we obtain an element $s \in \mathcal{F}(U)$ s.t. $s|_{U_x} = s_x$ for every $x \in X$. It remains to check that $\varphi_U(s) = t$. But this holds by 1.4(1) applied to $\mathcal{G}$, since by construction $\varphi_U(s)|_{U_x} = \varphi_{U_x}(s|_{U_x}) = \varphi_{U_x}(s_x) = t|_{U_x}$. Therefore, φ_U is surjective, hence an isomorphism. □

Definition 1.12 (Naive[1])**.** Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of presheaves. We say that φ is injective/surjective if for every $x \in X$, $\varphi_x : \mathcal{F}_x \to \mathcal{G}_x$ is injective/surjective. In particular, if $\mathcal{F}, \mathcal{G}$ are sheaves, then by proposition 1.11, φ is an isomorphism if and only if it is injective and surjective.

Definition 1.13. Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of presheaves on X. We define the following presheaves:

$$\begin{aligned} \ker(\varphi)(U) &:= \ker(\varphi_U), \\ \operatorname{im}(\varphi)^{\text{pre}}(U) &:= \operatorname{im}(\varphi_U) \\ \operatorname{coker}(\varphi)^{\text{pre}}(U) &:= \operatorname{coker}(\varphi_U) \end{aligned}$$

Remark 1.14. In each case, one easily checks that these assignments define presheaves. If $\mathcal{F}, \mathcal{G}$ are sheaves, then $\ker \varphi$ is a sheaf as well (Exercise), but in general $\operatorname{im} \varphi$ and $\operatorname{coker} \varphi$ are not sheaves.

Proposition 1.15 (Sheafification)**.** *Let $\mathcal{F}$ be a presheaf on X. Then there is a unique (up to unique isomorphism) sheaf $\mathcal{F}^+$, called* sheafification of $\mathcal{F}$*, and a morphism $\theta : \mathcal{F} \to \mathcal{F}^+$ with the following universal property: For any sheaf $\mathcal{G}$ and morphism $\varphi : \mathcal{F} \to \mathcal{G}$ of presheaves, there exists a unique morphism $\widetilde{\varphi} : \mathcal{F}^+ \to \mathcal{G}$ making the following diagram commute:*

$$\begin{tikzcd} \mathcal{F} \arrow[rr, "\varphi"] \arrow[dr, "\theta"'] & & \mathcal{G} \\ & \mathcal{F}^+ \arrow[ur, dashed, "\widetilde{\varphi}"'] & \end{tikzcd}$$

Moreover, θ induces isomorphisms $\theta_x : \mathcal{F}_x \to \mathcal{F}^+_x$ for every $x \in X$. If $\mathcal{F}$ is already a sheaf, then θ is an isomorphism.

Proof. We construct $\mathcal{F}^+$ as follows:

$$\mathcal{F}^+(U) := \left\{ s : U \to \coprod_{x \in U} \mathcal{F}_x \mid s(x) \in \mathcal{F}_x \text{ for all } x \in U, \text{ and } (*) \right\},$$

where $(*)$ is the following extra condition: For every $x \in U$ there exists an open neighbourhood $x \in V \subseteq U$ and $t \in \mathcal{F}(V)$ with $s(y) = t_y := [(V, t)]$ for all $y \in V$. Further, set $\theta_U(t)(x) = t_x$ for all $U \subseteq X$ open, $t \in \mathcal{F}(U)$ and $x \in U$. One checks that this $\mathcal{F}^+, \theta$ have the required properties:

It is clear that $\mathcal{F}^+$ is a sheaf, since it is constructed via functions with local properties, and that Lecture 3
θ is a morphism of presheaves, since $(t|_W)_x = t_x$ for $x \in W \subseteq V$, $t \in \mathcal{F}(V)$. Let $x \in X$ and Apr 20, 2026
$[(V, t)] \in \ker \theta_x$. Up to shrinking V, we have $\theta_V(t) = 0$. In particular, $[(V, t)] = \theta_V(t)(x) = 0$. Let $[(V, s)] \in \mathcal{F}^+_x$. By $(*)$ wlog there is $t \in \mathcal{F}(V)$ s.t. $s(y) = t_y$ for all $y \in V$. Hence $\theta_V(t) = [(V, s)]$. If $\mathcal{F}$ is a sheaf, then θ is thus an isomorphism by proposition 1.11.

Finally, we check the universal property: Let $\mathcal{G}$ be a sheaf, and $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of presheaves. By the previous paragraph, it suffices to give a morphism $\widetilde{\varphi} : \mathcal{F}^+ \to \mathcal{G}^+ \cong \mathcal{G}$. Let $U \subseteq X$ be open and let $\widetilde{\varphi}(s)(y) = \varphi_y(s(y))$. It is easy to check that this is a well-defined morphism of sheaves, makes the diagram commute, and is the only morphism with this property. The details are left as an exercise. □

[1] will be corrected in 1.21

Example 1.16. Let X be a topological space such that for every $x \in X$ there is a neighbourhood $x \in U_x \subsetneq X$, and let $0 \neq A$ an abelian group.

(i) Let $\mathcal{F}(X) = A$ and $\mathcal{F}(U) = 0$ otherwise. This is clearly not a sheaf. By the assumption on X we have $\mathcal{F}_x = 0$ for all $x \in X$, so $\mathcal{F}^+ = 0$.
(ii) Let $\mathcal{F}$ be the constant presheaf with value A as in remark 1.5. From the universal property one sees that $\mathcal{F}^+ = \underline{A}$ is the constant sheaf.

Definition 1.17. A sub-(pre)sheaf of a (pre)sheaf $\mathcal{F}$ on X is a (pre)sheaf $\mathcal{F}'$ s.t. for every open $U \subseteq X$ one has $\mathcal{F}'(U) \subseteq \mathcal{F}(U)$, and for every $V \subseteq U$ open, $\rho'_{UV} = \rho_{UV}|_{\mathcal{F}'(U)}$. In other words, the inclusions $\mathcal{F}'(U) \hookrightarrow \mathcal{F}(U)$ have to form a morphism of (pre)sheaves.

If $\mathcal{F}' \subseteq \mathcal{F}$ is a sub-presheaf, then for every $x \in X$ we have $\mathcal{F}'_x \subseteq \mathcal{F}_x$.

Proposition 1.18. *Let $\mathcal{F}$ be a sheaf and $\mathcal{F}'$ be a subsheaf. Then $\mathcal{F}' = \mathcal{F}$ if and only if $\mathcal{F}'_x = \mathcal{F}_x$ for all $x \in X$.*

Proof. Follows directly from proposition 1.11. □

Example 1.19. If $\varphi : \mathcal{F} \to \mathcal{G}$ is a morphism of presheaves, then $\ker \varphi$ and $\operatorname{im} \varphi^{\text{pre}}$ are sub-presheaves of $\mathcal{F}$ resp. $\mathcal{G}$.

Exercise 1.20. Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of presheaves. Show that $(\ker \varphi)_x = \ker(\varphi_x)$ and $(\operatorname{im} \varphi^{\text{pre}})_x = \operatorname{im}(\varphi_x)$.

Definition 1.21. Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of presheaves. We say that φ is injective if and only if $\ker \varphi = 0$. For presheaves, we say φ is surjective if $\operatorname{coker} \varphi^{\text{pre}} = 0$[2] If $\mathcal{F}, \mathcal{G}$ are sheaves, define the image and cokernel sheaves via $\operatorname{im} \varphi := (\operatorname{im} \varphi^{\text{pre}})^+$ and $\operatorname{coker} \varphi := (\operatorname{coker} \varphi^{\text{pre}})^+$. Then $\operatorname{im} \varphi$ can be identified with a subsheaf of $\mathcal{G}$ (see next proposition). We say that φ is surjective as a morphism of sheaves if $\operatorname{im} \varphi = G$, equivalently $\operatorname{coker} \varphi = 0$.

Proposition 1.22. *Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of sheaves. Then $\operatorname{im} \varphi$ can be seen as a subsheaf of $\mathcal{G}$. Moreover, $\operatorname{im} \varphi = \mathcal{G} \iff \varphi_x$ is surjective for all $x \in X$.* Lecture 4 Apr 23, 2026

Proof. We have $\operatorname{im} \varphi \subseteq \mathcal{G}^+ \cong \mathcal{G}$. The second part follows directly from proposition 1.11. □

Remark 1.23. Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of (pre)sheaves. Then φ is injective if and only if φ_U is injective for all $U \subseteq X$ open, that is $\ker \varphi = 0$. If φ is injective, then also $\varphi_x : \mathcal{F}_x \to \mathcal{G}_x$ is injective, but the converse is only true for sheaves.

φ is a surjective morphism of presheaves if φ_U is surjective for all $U \subseteq X$ open, that is $\operatorname{im} \varphi^{\text{pre}} = \mathcal{G}$. Again, if φ is surjective, then so are all φ_x, but the converse is not true in general (even for sheaves).

If $\mathcal{F}, \mathcal{G}$ are sheaves, φ is a surjective morphism of sheaves if and only if $\operatorname{im} \varphi = \mathcal{G}$, if and only if φ_x is surjective for all $x \in X$.

Example 1.24. (i) Let $\mathcal{F}, \mathcal{G}$ be (pre)sheaves on X. Then $(F \oplus \mathcal{G})(U) := \mathcal{F}(U) \oplus \mathcal{G}(U)$ is again a (pre)sheaf.
(ii) Let A be an abelian group, X a topological space and $x \in X$. The sheaf

$$ {}^x\mathcal{F}(U) := \begin{cases} A, & \text{if } x \in U, \\ 0 & \text{otherwise} \end{cases} $$

is called the *skyscraper sheaf*. One calculates ${}^x\mathcal{F}_y = A$ if $y \in \overline{\{x\}}$, and ${}^x\mathcal{F}_y = 0$ otherwise. In particular, if X is T_1 (that is, points are closed), then ${}^x\mathcal{F}_y \neq 0$ only for $x = y$.

[2] Note that this is not equivalent to φ_X being surjective for all $x \in X$.

(iii) Let X be a connected T_1-space, $x \neq z \in X$. Let $0 \neq A$ be an abelian group, and consider the natural map $\varphi : \underline{A} \to {}^x\mathcal{F} \oplus {}^z\mathcal{F}$. One sees that φ is surjective as a morphism of sheaves, but not as a morphism of presheaves.

Definition 1.25. Let $f : X \to Y$ be a continuous map of topological spaces.

(i) Let $\mathcal{F}$ be a sheaf on X. The *direct image* of $\mathcal{F}$ is the sheaf $f_*\mathcal{F}$ on Y defined by

$$f_*\mathcal{F}(U) := \mathcal{F}(f^{-1}(U)).$$

(ii) Let $\mathcal{G}$ be a sheaf on Y. The *inverse image* $f^{-1}\mathcal{G}$ of $\mathcal{G}$ is defined as the sheafification of the presheaf

$$U \mapsto \varinjlim_{V \supseteq f(U)} \mathcal{G}(V).$$

2 Affine Schemes

In Algebraic Geometry I, we showed an equivalence of categories between affine algebraic sets over K, and finitely generated reduced K-algebras, for K algebraically closed. We want to generalize this to allow any ring A on the algebraic side. A natural generalization of the classical setting is to consider $\operatorname{Spec} A$, and to endow this set with topological and algebraic information.

Let A be a ring (always commutative with 1).

Definition 2.1. $\operatorname{Spec} A := \{\mathfrak{p} \subseteq A \mid \mathfrak{p} \text{ prime ideal}\}$. If $I \subseteq A$ is any ideal, let $V(I) := \{\mathfrak{p} \in \operatorname{Spec} A \mid I \subseteq \mathfrak{p}\}$. (Note that there is a canonical bijection $V(I) \cong \operatorname{Spec} A/I$.) For $S \subseteq A$ any subset, we write $V(S) := V(\langle S \rangle)$.

Lemma 2.2. *Let $I, J \subseteq A$ be ideals.*

(i) $\sqrt{I} := \{f \in A \mid f^n \in I \text{ for some } n \geq 1\} = \bigcap_{\mathfrak{p} \in V(I)} \mathfrak{p}$.
(ii) $I \subseteq \sqrt{I}$ *and* $V(I) = V(\sqrt{I})$.
(iii) If $I \subseteq J$*, then* $V(J) \subseteq V(I)$.
(iv) If $V(J) \subseteq V(I)$ *if and only if* $\sqrt{I} \subseteq \sqrt{J}$.

Proof. (i) was proven in Commutative Algebra. (ii), (iii), and one direction of (iv) are clear. So assume $V(J) \subseteq V(I)$, then $\sqrt{I} = \bigcap_{\mathfrak{p} \in V(I)} \subseteq \bigcap_{\mathfrak{p} \in V(J)} = \sqrt{J}$ by (i). □

Lemma 2.3. *Let $I, J, \{I_\alpha\}_{\alpha \in I} \subseteq A$ be ideals.*

(i) $V(IJ) = V(I \cap J) = V(I) \cup V(J)$.
(ii) $V(\sum_\alpha I_\alpha) = \bigcap_\alpha V(I_\alpha)$.

Proof. (i) For a prime ideal $\mathfrak{p} \subseteq A$, we have $IJ \subseteq \mathfrak{p}$ if and only if $I \subseteq \mathfrak{p}$ or $J \subseteq \mathfrak{p}$, if and only if $I \cap J \subseteq \mathfrak{p}$.

(ii) Exercise. □

Definition 2.4. By lemma 2.3 (and the observations $V(A) = \emptyset$, $V(0) = \operatorname{Spec} A$), the collection $\{V(I) \mid I \subseteq A \text{ ideal}\}$ forms the set of closed sets of a topology on $\operatorname{Spec} A$, the so-called *Zariski topology*.

Remark 2.5. Not every point of $\operatorname{Spec} A$ is closed. More precisely, let $x \in \operatorname{Spec} A$. Then $\overline{\{x\}} = V(x)$, so $\{x\}$ is closed if and only if x is a maximal ideal.

Lecture 5
Apr 27, 2026

Example 2.6. (i) If A is an integral domain, then $0 \in \operatorname{Spec} A$ satisfies $\overline{\{0\}} = \operatorname{Spec} A$. In particular, every open in $\operatorname{Spec} A$ contains $\{0\}$. Such a point is called a *generic point* of $\operatorname{Spec} A$.

(ii) Let $\mathbb{A}^1 := \operatorname{Spec} K[x]$. Then $\mathbb{A}^1$ contains the generic point (0), plus closed points corresponding to normalised irreducible polynomials in $K[x]$. If K is algebraically closed, every irreducible polynomial is linear, so $\mathbb{A}^1$ looks very similar to affine space in the classical setting, the only difference being the additional generic point.

(iii) $\operatorname{Spec} \mathbb{Z} = \{(0)\} \cup \{(p) \mid p \text{ prime }\}$

(iv) Let $\mathbb{A}^2 := \operatorname{Spec} K[x, y]$ with K algebraically closed. It contains the generic point (0), closed points $(x - a, y - b)$ as in the classical settings, but also (principal) prime ideals corresponding to curves. Let (F) be such an ideal, then

$$\overline{\{(F)\}} = \{(F)\} \cup \{(x - a, y - b) \mid F(a, b) = 0\}.$$

Lemma 2.7. *(i) If A is an integral domain, then* $\operatorname{Spec} A$ *is irreducible. (The converse fails e.g. for* $K[x]/(x^2)$*). If A is reduced and* $\operatorname{Spec} A$ *is irreducible, then A is an integral domain.*

(ii) $\operatorname{Spec} A$ *is not connected if and only if there exists* $0, 1 \neq e \in A$ *with* $e^2 = e$*, if and only if* $A \cong A_1 \times A_2$ *with* $A_1, A_2 \neq 0$.

Proof. (i) Assume that $\operatorname{Spec} A$ is irreducible and A is reduced but not an integral domain. Let $a * b = 0$ with $0 \neq a, b \in A$. Then $\operatorname{Spec} A = V(0) = V(a) \cup V(b)$. If $V(a) = \operatorname{Spec} A$, then $a \in \sqrt{0} = 0$ by assumption, contradiction. Hence $\operatorname{Spec} A$ is reducible. The rest is left as an exercise. □

Definition 2.8. Let $f \in A$. Then $D(f) := \operatorname{Spec} A \setminus V(f) = \{\mathfrak{p} \in \operatorname{Spec} A \mid f \notin \mathfrak{p}\}$ is called a *distinguished* or *standard open* subset of $\operatorname{Spec} A$.

Remark 2.9. One has $D(f) \cap D(g) = D(fg)$. The set $\{D(f) \mid f \in A\}$ forms a basis of the Zariski topology on $\operatorname{Spec} A$: Let $U = \operatorname{Spec} A \setminus V(I)$ be open and $\mathfrak{p} \in U$. Since $\mathfrak{p} \notin V(I)$, $I \not\subseteq \mathfrak{p}$. Hence choose $f \in I \setminus \mathfrak{p}$, which satisfies $\mathfrak{p} \in D(f) \subseteq U$.

We want to construct a sheaf of "regular functions" on $\operatorname{Spec} A$. Recall from the classical setting, that if $X \subseteq \mathbb{A}^n$ was an affine algebraic set and $U \subseteq X$ open, we defined $\mathcal{O}_X(U)$ as the set of all functions $f : U \to K$ that locally are rational functions, and we proved $\mathcal{O}_X(D(f)) \cong A(X)_f$ and $\mathcal{O}_{X,x} \cong \varinjlim_{D(f) \ni x} A(X)_f \cong A(X)_{\mathfrak{m}_x}$. This motivates us to define a sheaf $\mathcal{O}_X$ of rings on $\operatorname{Spec} A$ s.t. $\mathcal{O}_X(D(f)) \cong A_f$. Note that if $D(g) \subseteq D(f)$, then $g^m = fa$ for some $a \in A, m \geq 1$ by lemma 2.2, so there is a natural map $A_f \to A_g$. Since we know what the stalks are supposed to look like, we can proceed as in the proof of sheafification 1.15.

Definition 2.10. Let $X = \operatorname{Spec} A$. The structure sheaf $\mathcal{O}_X$ is defined as

$$\mathcal{O}_X(U) := \left\{ s : U \to \coprod_{\mathfrak{p} \in U} A_\mathfrak{p} \mid s(\mathfrak{p}) \in A_\mathfrak{p} \text{ and } (*) \right\},$$

where $(*)$ is the extra condition that for every $\mathfrak{p} \in U$ there exists an open neighbourhood $\mathfrak{p} \in V \subseteq U$ and $a, f \in A$ such that $s(\mathfrak{q}) = \frac{a}{f}$ in $A_\mathfrak{q}$ for every $\mathfrak{q} \in V$. By construction it is clear that $\mathcal{O}_X$ is a sheaf of rings on $\operatorname{Spec} A$.

Proposition 2.11. *For* $\mathfrak{p} \in \operatorname{Spec} A$ *we have* $\mathcal{O}_{X,\mathfrak{p}} \cong A_\mathfrak{p}$*, and for* $f \in A$*,* $\mathcal{O}_X(D(f)) \cong A_f$*. In particular,* $\mathcal{O}_X(X) \cong A$.

Lecture 6
Apr 30, 2026

Proof. Let $\mathfrak{p} \in \operatorname{Spec} A$ and set $\varphi : \mathcal{O}_{X,\mathfrak{p}} \to A_\mathfrak{p}$, $[(U, f)] \mapsto f(\mathfrak{p})$. This is a well-defined homomorphism of rings. Let $\frac{a}{s} \in A_\mathfrak{p}$ with $a, s \in A$, $s \notin \mathfrak{p}$. That is, $\mathfrak{p} \in D(s)$, so $[(D(s), \mathfrak{q} \mapsto \frac{a}{s})]$ is a well-defined preimage of $\frac{a}{s}$. Conversely, let $[(U, f)] \in \ker \varphi$. By shrinking U, we may assume $f(\mathfrak{q}) = \frac{a}{s} \in A_\mathfrak{q}$, $a, s \in A$, for all $\mathfrak{q} \in U$. Then $0 = \varphi([(U, f)]) = \frac{a}{s} \in A_\mathfrak{p}$, hence $s'a = 0$ for some $s' \notin \mathfrak{p}$, i.e. $\rho \in D(s')$. By shrinking U further, if necessary, we may assume $U \subseteq D(s')$. Then $f = \frac{a}{s} = \frac{s'a}{s's} = 0$. Therefore, φ is an isomorphism $\mathcal{O}_{X,\mathfrak{p}} \cong A_\mathfrak{p}$.

Now let $f \in A$. Define $\psi : A_f \to \mathcal{O}_X(D(f))$, $\alpha \mapsto (t_\alpha : \mathfrak{q} \mapsto \alpha)$. This is clearly a well-defined homomorphism of rings. Let $\alpha = \frac{a}{f^n} \in \ker \psi$. That is, $\frac{a}{f^n} = 0$ in $A_\mathfrak{q}$ for every $\mathfrak{q} \in D(f)$. This means that for every $\mathfrak{q} \in D(f)$ there exists $s_\mathfrak{q} \notin \mathfrak{q}$ with $s_\mathfrak{q} a = 0$. Consider the ideal $I = \{b \in a \mid ba = 0\}$ (called the annihilator of a). Then for every $\mathfrak{q} \in D(f)$, $s_\mathfrak{q} \in I \setminus \mathfrak{q}$, i.e. $I \not\subseteq \mathfrak{q}$. Hence $V(I) \subseteq \operatorname{Spec} A \setminus D(f) = V(f)$. By lemma 2.2 it follows that $f^m \in I$. for some $m > 0$. Then $\alpha = \frac{a}{f^n} = \frac{af^m}{f^{n+m}} = 0$.

For surjectivity, let $t \in \mathcal{O}_X(D(f))$. By definition, there is an open cover $D(f) = \bigcup_i V_i$ s.t. $t|_{V_i} = \frac{a_i}{s_i}$ for some $a_i, s_i \in A$ with $V_i \subseteq D(s_i)$. By remark 2.9 wlog $V_i = D(h_i)$. Now $V(h_i) \supseteq V(s_i)$, so again by lemma 2.2, $h_i \in \sqrt{(s_i)}$, i.e. $h_i^{n_i} = s_i c_i$ for some $n_i > 0, c_i \in A$. It follows that $t|_{D(h_i)} = \frac{a_i}{s_i} = \frac{a_i c_i}{h_i^{n_i}}$. Replacing $h_i^{n_i}$ by h_i (since $D(h_i) = D(h_i^{n_i})$) and $a_i c_i$ by a_i, we simply write $t|_{D(h_i)} = \frac{a_i}{h_i}$.

Observe that there exists a finite subcover $D(f) = \bigcup_{i=1}^{k} D(h_i)$ of $\bigcup_i D(h_i)$. Indeed, $D(f) \subseteq \bigcup_i D(h_i)$ implies $V(f) \supseteq \bigcap_i V(h_i) = V(\sum_i (h_i))$ by lemma 2.3. By lemma 2.2, $f^n \in \sum_i (h_i)$ for some $n > 0$. But this means f^n is a *finite* linear combination of the h_i's. Taking only those h_i which appear in this sum and retracing the argument backwards yields the claim.

For $i \neq j$, we have $\frac{a_i h_j}{h_i h_j} = t|_{D(h_i)}|_{D(h_i h_j)} = t|_{D(h_j)}|_{D(h_i h_j)} = \frac{a_j h_i}{h_j h_i}$ as functions on $D(h_i h_j)$. By the injectivity of ψ (for $D(h_i h_j)$) proved above, even $\frac{a_i h_j}{h_i h_j} = \frac{a_j h_i}{h_j h_i}$ as elements of $A_{h_i h_j}$, i.e. $(h_i h_j)^n (a_i h_j - a_j h_i) = 0$ in A, for some n. Since there are only finitely many sets in the open cover, we can choose n big enough such that this equality holds for all i, j. Lecture 7 May 4, 2026

Replacing h_i by h_i^{n+1} and a_i by $h_i^n a_i$, we still have $t|_{D(h_i)} = \frac{a_i}{h_i}$, and now $h_j a_i = h_i a_j$ for all i, j. As before, $f \in \sqrt{(h_1, \ldots, h_k)}$, so we can write $f^m = \sum_i b_i h_i$ for some $m > 0$ and $b_i \in A$. Let $a := \sum_i b_i a_i$. We claim that $\psi(\frac{a}{f^m}) = t$. Indeed, for every j we have $f^m a_j = \sum_i b_i h_i a_j = \sum_i b_i a_i h_j = a h_j$, i.e. $\frac{a}{f^m} = \frac{a_j}{h_j} \in A_{\mathfrak{q}}$ for every $\mathfrak{q} \in D(h_j)$. $\square$

Having now associated to a ring A the pair $(\operatorname{Spec} A, \mathcal{O}_{\operatorname{Spec} A})$, we next want to define morphisms between these objects. We let us motivate by the following

Remark 2.12. Recall from classical algebraic geometry that if X, Y are quasi-projective algebraic sets, a map $\varphi : X \to Y$ was defined to be a morphism if φ is continuous, and for every open $U \subseteq Y$ and $f \in \mathcal{O}_Y(U)$, its preimage $\varphi^* f = f \circ \varphi$ is regular, i.e. lies in $\mathcal{O}_X(\varphi^{-1}(U))$. This yielded compatible homomorphisms of K-algebras $\mathcal{O}_Y(U) \to \mathcal{O}_X(\varphi^{-1}(U))$, in other words a morphism of sheaves $\mathcal{O}_Y \to \varphi_* \mathcal{O}_X$.

Definition 2.13.
(i) A *ringed space* is a pair $(X, \mathcal{O}_X)$, where X is a topological space and $\mathcal{O}_X$ a sheaf of rings on X.
(ii) A morphism of ringed spaces $(X, \mathcal{O}_X)$ and $(Y, \mathcal{O}_Y)$ is a pair $(f, f^\#)$, where $f : X \to Y$ is a continuous map, and $f^\# : \mathcal{O}_Y \to f_* \mathcal{O}_X$ is a morphism of sheaves.
(iii) A *locally ringed space* is a ringed space $(X, \mathcal{O}_X)$ such that all stalks $\mathcal{O}_{X,x}$, $x \in X$ are local rings.
(iv) A morphism of locally ringed spaces $(f, f^\#) : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ is a morphism of ringed spaces such that for every $x \in X$, the map $f^\#_x : \mathcal{O}_{Y, f(x)} \to \mathcal{O}_{X,x}$ is a local homomorphism of rings (where a morphism of rings $\varphi : (A, \mathfrak{m}) \to (B, \mathfrak{n})$ of local rings is called *local* if $\varphi^{-1}(\mathfrak{n}) = \mathfrak{m}$). The map $f^\#_x$ is defined as the compositum

$$\mathcal{O}_{Y,f(x)} \xrightarrow{f^\#_{f(x)}} (f_* \mathcal{O}_X)_{f(x)} = \varinjlim_{f^{-1}(U) \ni x} \mathcal{O}_X(f^{-1}(U)) \xrightarrow{\text{incl}_*} \varinjlim_{U \ni x} \mathcal{O}_X(U) = \mathcal{O}_{X,x},$$

where the second map is induced by inclusion of directed sets.

Remark 2.14. Let A be a ring. The pair $(\operatorname{Spec} A, \mathcal{O}_{\operatorname{Spec} A})$ is a locally ringed space by proposition 2.11.

Definition 2.15.
(i) Let $(f, f^\#) : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ and $(g, g^\#) : (Y, \mathcal{O}_Y) \to (Z, \mathcal{O}_Z)$ be morphisms of (locally) ringed spaces. Then their composition is defined as $(g, g^\#) \circ (f, f^\#) := (g \circ f, (g \circ f)^\#)$, with $(g \circ f)^\# := g_* f^\# \circ g^\# : \mathcal{O}_Z \to (g \circ f)_* \mathcal{O}_X$.
(ii) An isomorphism of (locally) ringed spaces is a morphism with a two-sided inverse morphism.

A morphism $(f, f^\#) : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ is an isomorphism if and only if $f : X \to Y$ is a homeomorphism and $f^\# : \mathcal{O}_Y \to f_* \mathcal{O}_X$ is an isomorphism of sheaves. (Exercise)

Lemma 2.16. *Let $\varphi : A \to B$ be a morphism of rings. Then $f : \operatorname{Spec} B \to \operatorname{Spec} A$, $\mathfrak{p} \mapsto \varphi^{-1}(\mathfrak{p})$ is continuous.*

Proof. f is well-defined since the preimage of a prime ideal is again prime. Let $V(I) \subseteq \operatorname{Spec} A$. Then $f^{-1}(V(I)) = \{\mathfrak{p} \in \operatorname{Spec} B \mid \varphi^{-1}(\mathfrak{p}) \in V(I)\} = V(\varphi(I))$ is closed. □

Lecture 8
May 7, 2026

Remark 2.17. Let $\varphi : A \to B$ be a ring morphism and $X = \operatorname{Spec} B, Y = \operatorname{Spec} A$. Then φ induces a morphism of locally ringed spaces $(f, f^\#) : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$, where f is the map of lemma 2.16, and $f^\# : \mathcal{O}_Y \to f_*\mathcal{O}_X$ is defined on $U \subseteq Y$ as

$$f^\#_U : \mathcal{O}_Y(U) \to \mathcal{O}_X(f^{-1}(U)), \quad t \mapsto (\mathfrak{p} \mapsto \varphi_\mathfrak{p}(t(f(\mathfrak{p})))),$$

where $\varphi_\mathfrak{p} : A_{f(\mathfrak{p})} \to B_\mathfrak{p}$ is the natural morphism induced by the universal property of localisation. Note that if $t|_V = \frac{a}{s}$, then $f^\#(t)|_{f^{-1}(V)} = \frac{\varphi(a)}{\varphi(s)}$, so $f^\#$ is indeed a well-defined morphism of sheaves.

We check that $(f, f^\#)$ is a morphism of locally ringed spaces, i.e. that $f^\#_\mathfrak{p} : \mathcal{O}_{Y,f(\mathfrak{p})} \to (f_*\mathcal{O}_X)_{f(\mathfrak{p})} \to \mathcal{O}_{X,\mathfrak{p}}$ is a local morphism of rings. By definition, this map is given by $[(U, t)] \mapsto [(f^{-1}(U), f^\#(t))]$, and hence by construction the diagram

$$\begin{tikzcd} \mathcal{O}_{Y,f(\mathfrak{p})} \arrow[r, "f^\#_\mathfrak{p}"] \arrow[d, "\cong"] & \mathcal{O}_{X,\mathfrak{p}} \arrow[d, "\cong"] \\ A_{f(\mathfrak{p})} \arrow[r, "\varphi_\mathfrak{p}"] & B_\mathfrak{p} \end{tikzcd}$$

commutes, where the vertical isomorphisms come from proposition 2.11. Since $\varphi_\mathfrak{p}$ is local, hence so is $f^\#_\mathfrak{p}$.

As was the case in classical algebraic geometry, we have

Proposition 2.18. *Let A, B be rings, and $X = \operatorname{Spec} B$, $Y = \operatorname{Spec} A$. Then the construction above yields a bijection*

$$\operatorname{Hom}_{\texttt{Ring}}(A, B) \xrightarrow{\cong} \operatorname{Hom}_{\texttt{LRS}}((X, \mathcal{O}_X), (Y, \mathcal{O}_Y)).$$

(In other words, the functor Spec *is fully faithful.)*

Proof. Let $(f, f^\#) : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ be a morphism of locally ringed spaces. Then $f^\#_Y : \mathcal{O}_Y(Y) \cong A \to B \cong f_*\mathcal{O}_X(Y)$, we claim that this construction is inverse to the one constructed in remark 2.17. Given $\varphi : A \to B$, let $(f, f^\#)$ be the induced morphism of locally ringed spaces. Then $f^\#_Y = \varphi$ (under the canoncial identifications) by construction.

Conversely, let $(f, f^\#)$ be a morphism of locally ringed spaces, and let $(g, g^\#)$ be the morphism construced from $\varphi := f^\#_Y$ as above. Let $\mathfrak{p} \in X$. We have a commutative diagram

$$\begin{tikzcd} B \cong \mathcal{O}_X(X) \arrow[d] & \mathcal{O}_Y(Y) \cong A \arrow[l, "f^\#_Y"'] \arrow[d] \\ B_\mathfrak{p} \cong \mathcal{O}_{X,\mathfrak{p}} & \mathcal{O}_{Y,f(\mathfrak{p})} \cong A_{f(\mathfrak{p})} \arrow[l, "f^\#_\mathfrak{p}"'] \end{tikzcd}$$

Since $f^\#_\mathfrak{p}$ is a local morphism, the preimage of $\mathfrak{p}B_\mathfrak{p}$ is the image of $f(\mathfrak{p})$ in $A_{f(\mathfrak{p})}$. By commutativity of the diagram, this equals $\varphi^{-1}(\mathfrak{p})$, i.e. $g = \varphi^{-1} = f$.

By the universal property of localizations, there is a unique map $A_{f(\mathfrak{p})} \to B_\mathfrak{p}$ making the above diagram commute. But both $f^\#_\mathfrak{p}$ and $\varphi_\mathfrak{p}$ do, so they are equal. By the construction in remark 2.17, this immediately implies $f^\# = g^\#$. □

Definition 2.19. An *affine scheme* is a locally ringed space isomorphic to $(\operatorname{Spec} A, \mathcal{O}_{\operatorname{Spec} A})$ for some ring A.

Remark 2.20. Immediately from the definition and proposition 2.18 it follows that the functor $\operatorname{Spec} : \texttt{Ring} \to \texttt{AffSch}$, $A \mapsto \operatorname{Spec} A$, $\varphi \mapsto (f, f^{\#})$ as in remark 2.17 is an (anti-)equivalence of categories.

Definition 2.21. A *scheme* is a locally ringed space $(X, \mathcal{O}_X)$ such that for every $x \in X$ there exists a neighbourhood $x \in U \subseteq X$ such that $(U, \mathcal{O}_X|_U)$ is an affine scheme.